SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO5 ASSESSMENT TOOL 1 – Research Paper Review / Literature Survey

Course Code:	CSA10 / CSA1063	**Course Title:**	Software Engineering
CO Assessed:	CO5	**Assessment:**	Assessment Tool 1 – **Research Paper Review / Literature Survey**
Weightage:	20%	**Total Marks:**	25
CO5 Statement:	Evaluate emerging technologies and societal impacts, maintaining and evolving software systems responsibly		
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Date:	24/08/2026	**Duration:**	60 minutes

Code of Conduct

I Dhanushree S/192421218 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action. Signature of the Student: Dhanushree S

Rubrics for Calculation

Evaluation Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Selection & Review of Research Papers(25%)	Selects highly relevant, recent, credible papers and provides comprehensive reviews of objectives, methodology, tools, datasets, and findings	Selects relevant papers and provides good reviews with minor gaps	Selects moderately relevant papers with basic reviews	Papers are irrelevant or reviews are incomplete

Comparative Analysis(20%)	Provides a detailed comparison of methodologies, tools, datasets, results, and performance using a clear comparison table	Provides a good comparison with relevant parameters	Provides limited comparison with few parameters	Little or no meaningful comparison
Critical Analysis &	Clearly analyzes limitations and identifies significant,	Identifies relevant limitations and research gaps	Identifies basic gaps with limited analysis	Research gaps are unclear or missing
Research Gap(25%)	well-supported research gaps			
Future Directions & Relevance(15 %)	Proposes practical and innovative future directions directly relevant to the assigned problem statement	Proposes relevant future directions	Provides general future suggestions	Future directions are missing or unrelated
Documentation , Citations & References(15 %)	Excellent academic structure, clear presentation, accurate citations, and consistent referencing	Well-organized with minor formatting/citation issues	Adequately documented with some errors	Poorly organized with inadequate or incorrect references

Criteria	**Mark**
Selection & Review of Research Papers(25%)	/5
Comparative Analysis(20%)	/5
Critical Analysis & Research Gap(25%)	/5
Future Directions & Relevance(15%)	/5
Documentation, Citations & References(15%)	/5
TOTAL	/25

Annexure A – Research Paper Review / Literature Survey

AI-BASED CROP DISEASE DETECTIONAND FARMER ADVISORY PLATFORM

1. Introduction

Agriculture depends heavily on timely identification and management of crop diseases. Diseases can reduce yield, lower product quality, increase production costs, and create financial uncertainty for farmers. Traditional diagnosis commonly depends on visual inspection and access to agricultural experts. These approaches can be slow, subjective, and difficult to scale when a farmer is working in a remote location. Computer vision and deep learning provide an opportunity to automate the first stage of disease recognition from images and connect the prediction to a farmer-oriented advisory service.

The assigned problem statement is AI-Based Crop Disease Detection and Farmer Advisory Platform. The proposed problem is broader than image classification alone because a useful platform should combine disease recognition, understandable recommendations, accessible software, and practical deployment constraints. Recent research therefore needs to be examined from several perspectives: model accuracy, dataset realism, computational cost, explainability, edge deployment, and farmer usability.

The supplied assessment document identifies five major evaluation dimensions: selection and review of research papers, comparative analysis, critical analysis and research gap, future directions and relevance, and documentation/citations/references. It also expects comprehensive review of objectives, methodologies, tools, datasets and findings. □filecite□turn0file0□L25-L36□

1.1 Objectives of the Literature Survey

• To study recent AI and deep-learning approaches for crop and plant disease detection.

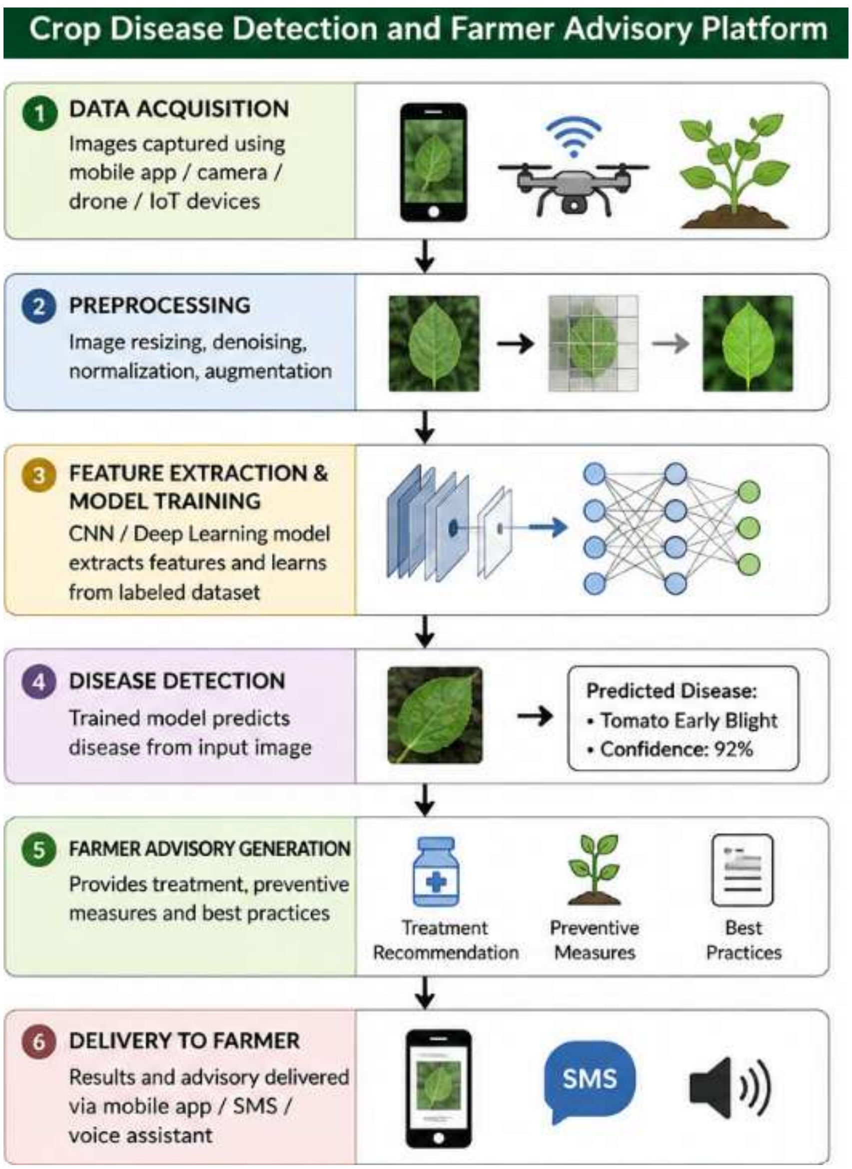
• To identify datasets, model architectures, preprocessing techniques and evaluation metrics used by researchers.

• To compare conventional deep learning, lightweight models, ensemble learning, explainable AI, drone-based detection and edge-computing approaches.

• To identify limitations that prevent laboratory-level results from translating directly into reliable field applications.

• To derive a research gap for a farmer advisory platform that is accurate, efficient, explainable and practical for rural use.

• To propose future improvements that connect disease recognition with actionable and accessible farmer support.



2. Background and Problem Definition

Crop disease detection is a computer-vision problem in which an input image is analyzed to determine whether a plant is healthy or affected by a particular disease. The task may be formulated as classification when one label is assigned to an image, detection when diseased regions or objects must be localized, or segmentation when the affected area must be separated at pixel level. A farmer advisory platform adds a second stage: the prediction must be converted into useful information such as likely disease name, confidence, preventive practices and appropriate next actions.

2.1 Role of Artificial Intelligence

Deep convolutional neural networks learn visual features such as edges, textures, spots, color changes and lesion patterns directly from training images. Transfer learning can reduce the amount of training required by starting from a model pretrained on a large image collection. Ensemble learning combines multiple models to improve robustness, while explainable AI can help users and developers understand which image regions contributed to a prediction.

2.2 Why a Platform is Needed

A standalone classifier gives only a predicted class. A farmer advisory platform should provide an end-to-end workflow: image capture, quality checking, disease prediction, confidence estimation, explanation, advisory retrieval and user feedback. It should also consider language accessibility and limited connectivity. Recent work demonstrates separate pieces of this pipeline, but the literature indicates that strong benchmark accuracy does not automatically guarantee reliable field deployment.

2.3 Core Workflow

A practical conceptual workflow is: Farmer → Mobile/Web Interface → Image Capture → Image Quality Check → Preprocessing → AI Disease Classifier → Confidence/Explainability → Disease Knowledge Base → Farmer Advisory → Feedback and Continuous Improvement.

3. Literature Search Strategy

The survey focuses mainly on recent research from 2022–2026, while using the PlantVillage benchmark background where necessary because it remains central to the literature. The search considered peer-reviewed journal and conference-style publications and scholarly technical sources. The selected studies represent different research directions rather than repeating the same architecture.

3.1 Selection Criteria

• Direct relevance to plant/crop disease detection or farmer-oriented agricultural AI.

• Use of measurable evaluation such as accuracy, precision, recall or F1-score.

• Clear description of model methodology and dataset.

• Evidence of deployment considerations such as drones, edge devices, lightweight networks or explainability.

• Recent publication date where possible.

• Potential contribution to the assigned farmer advisory platform.

3.2 Selected Research Directions

Six major studies/directions are reviewed in detail: an improved Efficient CNN drone system (2022), a deep-learning plant disease model using PlantVillage and AI-Hub data (2023), an edge-computing approach for smallholder farming (2023), lightweight deep learning for smart monitoring (2024), ensemble learning with explainable AI (2024), and a hybrid Inception-Xception approach using multiple disease datasets (2025). A 2026 edge-aware comparative evaluation is also used as a very recent deployment-oriented reference.

4. Review of Paper 1 – AI-Based Drone Plant Disease Detection

Paper: Artificial Intelligence-Based Drone System for Multiclass Plant Disease Detection Using an Improved Efficient Convolutional Neural Network, Frontiers in Plant Science, 2022. The study evaluates plant disease classification using the PlantVillage dataset and additionally validates the idea with real images captured using a DJI Phantom 3 drone under natural lighting. The PlantVillage collection used in the study contains tens of thousands of images across multiple plant species and disease classes. □cite□turn0search4□

4.1 Objective

The main objective is to improve multiclass plant disease detection using an efficient convolutional neural network and extend the evaluation beyond controlled leaf images by using drone-captured samples.

4.2 Methodology

The approach uses an improved efficient CNN for classification. Standard metrics such as accuracy, precision, recall and F1-score are used for evaluation. The important contribution for the present problem is the attempt to connect AI classification with aerial data acquisition, which moves the application closer to agricultural field monitoring. □cite□turn0search4□

4.3 Dataset and Tools

PlantVillage is the primary benchmark. The authors also collected approximately 30 colored samples for selected species using a DJI Phantom 3 Standard quadcopter, flying at approximately 5–6 m altitude. This natural-light validation is valuable because real field images differ from controlled laboratory-style images. □cite□turn0search4□

4.4 Findings and Limitation

The study demonstrates that deep-learning disease classification can be combined with drone imagery. However, the real-world sample collection is relatively small compared with the benchmark dataset, and the approach is more focused on detection/classification than on the complete farmer advisory experience. Drone hardware, flight operation, image quality and cost can also limit widespread use.

Figure 2. Healthy vs Diseased Leaves

5. Review of Paper 2 – Deep Learning Plant Disease Model

Paper: Construction of deep learning-based disease detection model in plants, Scientific Reports, 2023. The study reports use of the PlantVillage dataset together with the AI-Hub dataset, demonstrating an effort to evaluate plant disease recognition using more than one source. □cite□turn0search8□

5.1 Objective and Methodology

The research aims to construct a deep-learning model for plant disease detection. The use of established image datasets supports controlled experimentation and comparison of model performance. The study is relevant because it illustrates the common research pattern of training on benchmark datasets and validating disease recognition using additional data.

5.2 Dataset

The paper states that the PlantVillage dataset is available from its original repository and that the AI-Hub dataset is also used. This combination is important because a second dataset can expose differences in image distribution and reduce dependence on one benchmark source. □cite□turn0search8□

5.3 Critical Review

A major strength is the use of multiple data sources. A continuing challenge, however, is domain shift: images collected in laboratories or curated datasets can differ from photographs taken by farmers using different phones, lighting conditions, backgrounds and disease stages. A farmer advisory platform should therefore evaluate not only overall accuracy but also robustness across crops, devices and environments.

6. Review of Paper 3 – Edge Computing for Smallholder Farming

Paper: AI-Powered Crop Disease Detection Using Edge Computing for Smallholder Farming Systems, National Journal of Smart Agriculture and Rural Innovation, 2023. The paper proposes a crop disease detection system designed for low-cost edge computing devices and offline rural environments. It explicitly considers smallholder agriculture and real-time classification. □cite□turn0search36□

6.1 Objective

The objective is to make AI-based disease detection available where continuous internet connectivity and high-end computing resources may not be available.

6.2 Methodology and Technology

The approach emphasizes convolutional neural networks deployed on low-cost edge hardware such as Raspberry Pi. The design direction is significant for the assigned platform because inference can happen near the user instead of sending every image to a cloud server.

6.3 Strengths

• Supports offline or low-connectivity scenarios.

• Reduces dependence on cloud inference.

• Can improve response time for local predictions.

• Targets affordability and smallholder farming constraints.

6.4 Limitations

Edge devices have limited CPU, memory and power compared with cloud GPUs. Therefore, a model that is accurate but computationally heavy may not be practical. Compression, quantization, pruning and lightweight architectures become important. The paper's deployment perspective should be combined with stronger field datasets and farmer-centered advisory design.

7. Review of Paper 4 – Lightweight Deep Learning for Smart Monitoring

Paper: Grow-light smart monitoring system leveraging lightweight deep learning for plant disease classification, Artificial Intelligence in Agriculture, 2024. The work investigates lightweight adaptations of modern computer-vision architectures to retain performance while reducing computational requirements. □cite□turn0search10□

7.1 Methodology

The study develops lightweight adaptations involving inception blocks, residual connections and dense residual connections, and evaluates them on PlantVillage. The dataset description notes 54,306 images across 38 classes when background images are excluded, with 14 plant species represented. □cite□turn0search10□

7.2 Relevance

The key lesson is that accuracy alone is not sufficient for a deployable platform. A farmer-facing system may run on a smartphone, Raspberry Pi or another edge device, so inference time, memory footprint and energy consumption should be treated as first-class metrics.

7.3 Limitation

PlantVillage images largely contain isolated leaves with relatively simple backgrounds. Therefore, lightweight performance measured on this benchmark should not be assumed to represent performance on complex field scenes. Field photographs may contain multiple leaves, soil, shadows, insects, blur and occlusion.

8. Review of Paper 5 – Ensemble Learning and Explainable AI

Paper: Plant Leaf Disease Detection Using Ensemble Learning and Explainable AI, IEEE Access, 2024. The study combines several deep-learning models through ensemble learning and uses LIME for interpretation. The reported methodology includes VGG16, VGG19, InceptionV3 and ResNet101V2, with soft and hard voting. □cite□turn0search5□

8.1 Methodology

The workflow consists of data collection, preprocessing, model training/testing, ensemble voting and interpretation. Preprocessing includes scaling and augmentation operations such as rotation, zoom, shear and horizontal flipping. LIME is used to provide an explanation of the prediction. □cite□turn0search5□

8.2 Importance of Explainability

For a farmer advisory platform, explainability can improve trust. A system can highlight the region of the leaf that influenced the classification, helping the user distinguish a meaningful disease symptom from an irrelevant background. Explainability also helps developers identify failure cases and spurious model behavior.

8.3 Limitation

Ensembling several large networks increases computational cost. The approach therefore creates a trade-off between predictive performance and deployment efficiency. LIME explanations also need careful interpretation because an explanation is not equivalent to biological proof of disease.

9. Review of Paper 6 – Hybrid Inception-Xception Disease Classification

Paper: A novel hybrid inception-xception convolutional neural network for efficient plant disease classification and detection, Scientific Reports, 2025. The research considers several datasets, including PlantVillage, PlantDoc, a Turkey Plant Disease dataset and a Rice Disease dataset. PlantDoc is described as containing 2,569 images from 13 plant species and 30 classes, while the rice dataset contains 4,684 images covering bacterial blight, brown spot and leaf smut. citeturn0search12

9.1 Contribution

The use of multiple datasets is valuable because it moves evaluation beyond one benchmark. Different datasets contain different crops, backgrounds, camera conditions and disease appearances. A hybrid architecture also explores whether complementary feature extraction components can improve disease classification.

9.2 Critical Analysis

Multiple datasets increase diversity but do not automatically guarantee unbiased generalization. Dataset size, class balance, collection conditions and label quality remain important. A future farmer platform should include locally collected images and test data that are completely separated from training images.

10. Review of Paper 7 – Recent Edge-Aware Comparative Evaluation

A 2026 PLOS One study compares deep-learning architectures for plant disease classification with an explicit edge-performance perspective. It uses a 15-class subset of PlantVillage containing 20,639 images and evaluates models in the context of a robotic plant disease detection system. citeturn0search11

10.1 Relevance

This direction directly supports the idea that deployment metrics should accompany classification metrics. A practical system should report not only accuracy but also inference time, memory requirements, model size and hardware performance.

10.2 Implication

For the assigned platform, model selection should be treated as a multi-objective decision. The highest-accuracy architecture may not be the best choice if it requires expensive hardware or has unacceptable response time. A slightly less accurate lightweight model may be preferable when it enables offline, low-cost and fast inference.

11. Dataset Analysis

PlantVillage remains one of the most widely used datasets in plant disease research. Its public repository describes 54,306 images of healthy and diseased leaves across 14 crop species and 26 diseases in the original benchmark description. The dataset was introduced to support automated plant disease diagnosis, including smartphone-oriented applications. □cite□turn0search0□

11.1 Advantages

• Large and publicly accessible.

• Many crop/disease categories.

• Useful for reproducible benchmarking.

• Supports rapid comparison of architectures.

11.2 Limitations

• Controlled backgrounds can make classification easier than field photography.

• Disease appearance may be represented by limited stages or conditions.

• Class distributions may be unequal.

• Benchmark performance can overestimate real-world generalization.

• It is primarily an image classification resource rather than a complete advisory dataset.

11.3 Additional Datasets

PlantDoc and crop-specific datasets can complement PlantVillage. The literature reviewed here indicates that PlantDoc has substantially fewer images than PlantVillage but contains more field-like variation. Crop-specific rice and regional datasets can add local relevance. □cite□turn0search12□

12. Comparative Literature Survey

Study / Year	Methodology	Dataset / Tools	Main Contribution	Main Limitation

Drone + Efficient CNN (2022)	Improved efficient CNN; aerial validation	PlantVillage + DJI drone images	Connects AI classification with drone-based field monitoring	Limited real-world sample; hardware dependence
Deep-learning model (2023)	Deep learning classification	PlantVillage + AI-Hub	Uses more than one dataset	Domain shift and field generalization remain challenges
Edge computing (2023)	CNN on low-cost edge hardware	Rural/edge deployment; Raspberry Pi direction	Offline, low-cost, real-time concept	Resource limits can restrict model complexity
Lightweight DL (2024)	Lightweight inception/residual designs	PlantVillage	Reduces computational requirements	Benchmark images may be simpler than field scenes
Ensemble + XAI (2024)	VGG16/VGG19/InceptionV3/ResNet101V2 + voting + LIME	PlantVillage	Improves robustness and adds interpretation	High computational cost; explanations need careful interpretation
Hybrid Inception-	Hybrid CNN	PlantVillage, PlantDoc, Turkey and	Broader multi-dataset evaluation	Dataset differences and class

Xception (2025)		Rice datasets		balance remain issues
Edge-aware comparison (2026)	Comparative DL evaluation with hardware perspective	PlantVillage subset; robotic/edge setting	Emphasizes deployment performance	Benchmark dependence remains a concern

13. Comparative Analysis

The literature shows a clear evolution from high-performing image classifiers toward systems that consider deployment. Early benchmark-driven approaches emphasize accuracy. Later work increasingly investigates lightweight architectures, edge computing, explainability and multi-dataset evaluation.

13.1 Accuracy vs Generalization

High benchmark accuracy is useful but cannot alone establish field reliability. The difference between controlled leaf photographs and farmer-captured images is a major source of uncertainty. Therefore, the proposed platform should include a field-test dataset collected under different phones, lighting conditions, backgrounds, disease severities and geographic settings.

13.2 Performance vs Efficiency

Large CNNs and ensembles can deliver strong representation capacity but may be expensive to run. Lightweight architectures and edge inference reduce latency and connectivity requirements. The best solution is therefore likely to use a compact model for routine inference and optional cloud processing for difficult or low-confidence cases.

13.3 Explainability vs Complexity

Explainability is valuable for trust and debugging, but it adds processing and interface complexity. A practical system can use a simple visual explanation, such as highlighting the suspected lesion area, while presenting confidence and a short advisory rather than overwhelming the farmer with technical details.

14. Critical Analysis of Existing Approaches

The studies collectively demonstrate strong progress in AI-based plant disease recognition, but several unresolved challenges remain. First, benchmark datasets do not fully represent real agricultural environments. Second, model accuracy is often emphasized more strongly than latency, memory, energy use and user experience. Third, disease prediction and advisory generation are often treated as separate problems.

14.1 Technical Challenges

• Background clutter and multiple leaves in one image.

• Blur, shadows, poor lighting and camera differences.

• Similar visual symptoms across diseases.

• Imbalanced disease classes.

• Limited regional and crop-specific training data.

• High computational cost of large models.

• Uncertainty when the image does not contain enough evidence.

14.2 Software Engineering Challenges

From a software-engineering perspective, the platform should be modular and maintainable. The image-processing component, model service, advisory knowledge base, user interface and feedback mechanism should be separated so that one component can evolve without rewriting the entire application. Responsible maintenance also requires monitoring model drift, updating disease information and protecting user data.

15. Identified Research Gap

The strongest research gap identified from the reviewed literature is the lack of a single, practical, farmer-centered system that jointly optimizes disease-recognition accuracy, real-world robustness, computational efficiency, explainability and actionable multilingual advisory support.

15.1 Gap 1 – Real-World Generalization

Many experiments depend heavily on PlantVillage or similarly curated datasets. There is a need for larger field datasets captured using ordinary smartphones under natural conditions.

15.2 Gap 2 – Integrated Advisory

Disease classification does not automatically provide safe and useful guidance. A knowledge layer is required to map predicted disease and crop context to understandable management recommendations.

15.3 Gap 3 – Edge and Cloud Balance

Edge-only systems may be constrained by hardware, while cloud-only systems depend on connectivity. A hybrid architecture that performs normal inference locally and escalates uncertain cases to the cloud is a promising direction.

15.4 Gap 4 – Explainable and Trustworthy Prediction

Farmers and agricultural professionals may hesitate to act on an opaque prediction. The system should communicate confidence, show the relevant image region and clearly indicate when expert verification is recommended.

15.5 Gap 5 – Multilingual Accessibility

A technically accurate model may have limited practical value if its output is not understandable to the target users. A farmer platform should support regional languages and simple advisory messages.

16. Proposed Improved System

Based on the literature, an improved platform can be designed as a modular AI-assisted advisory system. The farmer captures a crop image using a mobile device. The application performs image-quality checks and preprocessing. A lightweight CNN or efficient vision model performs local inference. If confidence is high, the system immediately displays the disease and advisory. If confidence is low, the image can be sent to a cloud model or flagged for expert review.

16.1 Proposed Architecture

Farmer → Mobile/Web App → Image Quality Check → Preprocessing → Lightweight AI Model → Confidence Estimation → Explainability Module → Advisory Knowledge Base → Multilingual Recommendation → Feedback Store.

16.2 Advisory Layer

The advisory layer should not simply generate unrestricted medical-like recommendations. It should use a curated agricultural knowledge base containing disease symptoms, preventive practices, cultural management, safe handling information and escalation guidance. The prediction should be presented as an AI-assisted identification rather than an unquestionable diagnosis.

16.3 Feedback and Continuous Learning

User feedback can identify incorrect predictions and difficult cases. With appropriate privacy and validation controls, verified examples can be added to future training cycles. This creates a continuous improvement loop and reduces the risk that the model remains tied to an outdated dataset.

17. Future Research Directions

17.1 Larger Field Dataset

Build a geographically diverse dataset collected from farms using ordinary smartphones. Images should include natural backgrounds, different lighting conditions, disease stages and multiple camera qualities.

17.2 Lightweight and Compressed Models

Investigate pruning, quantization, knowledge distillation and efficient CNN/vision architectures so that inference can run on affordable mobile or edge hardware.

17.3 Hybrid Edge–Cloud Architecture

Use local inference for speed and offline operation while using cloud inference for uncertain or difficult cases. This provides a practical balance between availability and model capacity.

17.4 Explainable AI

Provide visual evidence, confidence scores and concise explanations. Compare explanation methods for reliability and usability rather than assuming that any heatmap is automatically trustworthy.

17.5 Multilingual Advisory

Support regional languages and voice-assisted interaction so that farmers can receive recommendations in a familiar form.

17.6 Multi-Modal Decision Support

Combine leaf images with crop type, growth stage, location, weather and farmer-entered symptoms where appropriate. This can reduce errors caused by visually similar diseases.

17.7 Privacy and Responsible AI

Minimize collection of personal information, protect uploaded images, document model limitations and provide an escalation path to agricultural experts.

17.8 Field Validation

Evaluate the final system on previously unseen farms and devices. Report not only accuracy but precision, recall, F1-score, inference time, model size, energy consumption and user satisfaction.

18. Expected Performance Evaluation Framework

The proposed platform should be evaluated using a multi-dimensional framework rather than a single accuracy number.

• Classification: accuracy, precision, recall, F1-score and confusion matrix.

• Robustness: performance on field images, unseen devices and unseen locations.

• Efficiency: inference time, memory usage, model size and energy consumption.

• Usability: task completion time, readability and farmer feedback.

• Explainability: localization quality and user understanding of the explanation.

• Advisory usefulness: correctness, relevance and clarity of recommendations.

18.1 Suggested Experimental Comparison

A fair experiment can compare a conventional CNN baseline, a transfer-learning model, a lightweight model and an ensemble model. The same test set should be used for all models. A second field test set should remain completely separate from training and validation data. The final model should be selected using a weighted decision that considers accuracy and deployment cost.

19. Overall Findings and Conclusion

The literature survey demonstrates that artificial intelligence has become a strong approach for automated crop disease recognition. Deep CNNs, efficient architectures, ensemble learning, explainable AI, drone imaging and edge computing each address different parts of the problem. PlantVillage provides a valuable benchmark and has supported extensive research, but benchmark results should not be interpreted as complete evidence of field readiness. □cite□turn0search0□turn0search10□

The main conclusion is that the assigned AI-Based Crop Disease Detection and Farmer Advisory Platform should not be designed as a simple classifier. A more useful system combines a lightweight and accurate model, real-world field data, confidence estimation, explainability, an advisory knowledge base, multilingual interaction and an edge/cloud deployment strategy.

The research gap therefore lies in integration and real-world validation. Future research should focus on building a complete and maintainable platform that can work under practical agricultural conditions while clearly communicating uncertainty. Such a system would better align AI performance with farmer needs and the Software Engineering CO5 emphasis on responsibly maintaining and evolving software systems.

20. References

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